

Harsh Pant

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Experience

Software Engineer, PharynxAI

July 2024 – Present

- Engineered **FastAPI and Flask backend services** integrated with **RunPod serverless pods**, deploying AI inference workflows as **Dockerized microservices**; implemented **Win32Pipe IPC** for low-latency cross-process communication between concurrent AI services.
- Built a **real-time multi-service data pipeline** using **LiveKit and WebRTC**, orchestrating STT, LLM, TTS, and avatar generation services end-to-end; developed **internal analytics tooling** to monitor session-level metrics (participant counts, session duration) across distributed deployments.
- Built an **automatic annotation pipeline** for internal use combining SeC and SAM2 for zero-shot segmentation and labeling; deployed alongside a **YOLO and RF-DETR detection pipeline** trained on **300K+ annotated images**, with systematic benchmarking to optimize detection and segmentation accuracy for production evaluation.
- Conducted **applied R&D**, translating research papers into production implementations with structured experimentation, iterative benchmarking, and parallelized inference pipelines for scalable generative AI workloads.

AI/ML Intern, Deepmindz Innovations

Jan 2024 – June 2024

- Built and analyzed **end-to-end ML pipelines on multi-modal datasets** (audio, text, and tabular), performing **EDA, feature engineering, and model evaluation** to benchmark custom-trained models against state-of-the-art methods.
- Developed **backend services and REST APIs** for AI workflow orchestration, integrating data preprocessing, model inference, and post-processing stages into deployable pipelines.
- Fine-tuned **LLMs and diffusion models** (LoRA-based Stable Diffusion), establishing **systematic evaluation benchmarks** to measure and drive iterative performance improvements.

Projects

Interview Assist

[Project Link](#)

- Built a **full-stack SaaS recruitment platform** (React + Flask) with LLM-powered screening, reducing candidate evaluation time by **60%** through RAG pipelines, few-shot prompting, and structured querying.
- Engineered a **LangChain RAG architecture** with document parsing, embedding generation, and vector database semantic search; exposed via **REST APIs with MongoDB-backed storage** for recruiter analytics and candidate insights.
- Implemented **RBAC-based access control** and production-ready API orchestration for reliable multi-tenant LLM service management.

Churn Prediction Model

[Project Link](#)

- Performed **EDA and feature engineering** on a customer churn dataset, training and evaluating models including RandomForest, Logistic Regression, AutoGluon, and Neural Networks.
- Benchmarked model performance, identified feature limitations through **systematic analysis**, and recommended additional data sources to improve predictive accuracy.

Education

Gautam Buddha University, Bachelor of Technology in Artificial Intelligence

Sept. 2021 – July 2025

- **CGPA:** 8.8
- **Coursework:** Data Structures and Algorithms, Database Management Systems, Operating Systems, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Large Language Models (LLMs)

Technologies

Languages: Python, JavaScript, R

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, WebRTC, LiveKit

Frontend: React, Tailwind CSS

Data & ML: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, HuggingFace, Transformers

AI Tools: LangChain, OpenCV, Stable Diffusion, YOLO

Databases: MongoDB, SQL

DevOps & Cloud: Docker, Git, GitHub, Linux (Ubuntu, Arch), AWS, Google Cloud Platform, Azure, RunPod

Other: Data Visualization (Matplotlib, Seaborn), IPC (Win32Pipe), MLOps